26/02/2026 DBMS Assignment - 01

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Question1: What are the Key reasons why a DBMS is preferable to a traditional file system? How does it ensure data security & consistency?

Scenario5: A ~~hospital~~ hospital wants to store, patient, doctor and appointment data in a DBMS.

Ques2: Identify the entities, attributes and relationships in the system. Draw the ER diagram and explain the key designs decisions you made.

Scenario6: An online shopping platform needs to store data about customers, orders and products.

Ques3: Suggest primary keys, foriegn keys, and integrity constraints for the system. How do these constraints help maintain the accuracy of data?

Ques4: Why is understanding Entity-Relationship models important when designing a DB? What could happen if a relationship is not correctly defined between entities?

Answers

Ans1 DBMS is preffered because it offers centralized control, reduced data redundancy and improved data structure & integrity, security and concurrency. DBMS provides ACID-compliant environment, efficient data retrieval and insertion.

Key reasons:

→ Impoved data security. → Data indepence
→ Data integrity & accuracy → Efficient data sharing, retrival & insertion

DBMS secures data security because database admins assign specific user roles (read, write, delete), ensuring that only authorized users can alter database. Data is protected through encryption and authentication maintaining secure environment.

Transactions in DBMS follow Atomic, Consistent, Isolated and Durable properties ensuring that even when a system crashes, data remains consistent.

Constraints prevent invalid data inputs ensuring proper Data validations.

Ans 2: Core entities are : Patient, Doctor and Appointment where Appointment is an associative entity linking patients and doctors. Key attributes can be unique IDs (Primary Key), names, specialities and appointment dates.

→ ~~Entities~~ Entities & Attributes:

for patient:- Patient ID (PK), names, DOB, address, phone.
for doctor:- Doctor ID (PK), Department, speciality.
for appointment:- Date, time, reason, Patient ID (FK),

→ Relationships :

Patient ↔ Doctor (many to many via appointment).
Appointment connects Patient & Doctor (one-to many)

Ans 3: Primary Keys : A column that uniquely identifies each record in a table. It enforces the entity integrity constraint, requiring that every record has a unique, non-null identifier.

Foreign Keys: A column in one table that refers to the primary key of another table, making a link between them. It enforces the referential integrity constraint.

Integrity constraints :

→ Unique cons. :- Ensuring all values in a column are unique, similar to primary key, but allows null values.

→ Not null cons :- Ensures a column cannot contain a NULL value.

→ Check cons :- Enforce that all values in a column satisfy a specific condition or range.

→ Default cons :- Automatically assigns a ~~the~~ default value to a column if no value is explicitly assigned.

They help maintain data accuracy. because they prevent invalid entries, ensure uniqueness, maintain logical relationships and improve consistency.